ANKITA KUMARI SATAPATHY

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PROFILE SUMMARY

Final-year Electrical and Electronics Engineering undergraduate at NIST University. Completed research internship experience at IIT Hyderabad. Experienced in MATLAB/Simulink-based modelling of grid-connected renewable-energy systems, power electronics, inverter control, and MPPT through technical projects. Seeking an entry-level engineering role in Solar PV, power systems, or renewable-energy integration.

EDUCATION

B.Tech, Electrical & Electronics Engineering — NIST University, Berhampur, Odisha *Sept 2023 – Present*
CGPA: 9.81 / 10 (Till 6th Semester)

Class XII (PCM), CBSE — PM Shri Kendriya Vidyalaya No. 2, AFS Kalaikunda, West Bengal 2023 | 90.4%

Class X, CBSE — PM Shri Kendriya Vidyalaya No. 2, AFS Kalaikunda, West Bengal 2021 | 94%

TECHNICAL SKILLS

Power & Energy: Power Systems, Renewable Energy Systems, Electrical Machines, Power Electronics, Control Systems
Renewable Energy & Grid Integration: Solar PV System Modelling, MPPT, Grid-Connected Inverter Control, PLL Synchronization
Simulation Tools: MATLAB, Simulink, ETAP
Programming: C, Python
Additional Skills: Electrical Circuit Analysis, System Modelling, Technical Documentation

INTERNSHIPS

Research Intern — Summer Undergraduate Research Exposure (SURE) 2026, IIT Hyderabad *May – Jul 2026*

• Selected for the IIT Hyderabad SURE research program; worked on renewable energy system modelling, power electronics and grid-synchronization techniques using MATLAB/Simulink.

Student Intern — Very Large-Scale Integration (VLSI), CTTC (MSME), Govt. of India, Bhubaneswar *Jun – Jul 2025*

• Gained hands-on experience in digital/VLSI design using Verilog HDL, Xilinx ISE, Quartus, DSCH2, and Microwind, including HDL modelling, simulation, and CMOS layout.

Student Intern — MATLAB & its application to Engineering, NIST University, Berhampur *Jul – Aug 2024*

• Developed MATLAB GUI applications and performed data analysis, visualization, and engineering simulations.
• Applied MATLAB to signal processing, image processing, and numerical analysis workflows.

PROJECTS

Hybrid Renewable Energy System with Intelligent Energy Management System *Aug 2026 – Present*

• Developing a hybrid renewable-energy system concept integrating solar PV, power electronic converters, and intelligent energy management.
• Studying power flow between renewable generation, energy storage, and electrical loads to address generation–demand mismatch.

Design, Control and Implementation of Grid-Connected Renewable Energy System *May 2026 – Jul 2026*

• Developed and analyzed grid-connected electrical system models in MATLAB/Simulink, including three-phase VSI, current/DC-link control, filters, boost converter, and P&O MPPT.
• Designed and evaluated SRF-PLL and SOGI-PLL for grid synchronization and implemented PLL algorithms on Arduino Due for experimental validation under different operating conditions.

BLDC Motor Speed Control Using Different Controllers *Dec 2025 – Sept 2026*

• Developed a BLDC motor drive model in MATLAB/Simulink for closed-loop speed control and analyzed motor response under different operating conditions.
• Implemented and compared PI, Fuzzy-PI, and ANN-based controllers for motor speed regulation under varying load conditions.

Power System Analysis Using ETAP *April 2026 – May 2026*

• Developed an electrical network model in ETAP and performed load-flow analysis to evaluate bus voltages, power flows, and system loading.
• Performed short-circuit analysis and interpreted fault-current levels under different loading conditions.

CERTIFICATIONS

- NPTEL – Smart Grid: Basics to Advanced Technologies — Elite + Silver (80%), 2026
- Summer Undergraduate Research Exposure (SURE) Internship, IIT Hyderabad (2026)
- Very Large-Scale Integration, CTTC (MSME), Government of India (2025)
- MATLAB and Its Application in Engineering, NIST University (2024)
- Electric Vehicle Workshop, Pantech e-Learning (2025)